MILITARY SPECIFICATION

MOUNT, TELESCOPE: M128A1

1. SCOPE

1.1 This specification covers performance requirements, related quality assurance provisions, and packaging data for a telescope mount which supports Elbow Telescope: M109 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWING

U.S. Army, Frankford Arsenal

F10559699	Mount, Telescope: M128A1

PACKAGING DATA SHEET

10559699	Packaging of Mount, Telescope: M128A1

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication.- The mount shall be manufactured in accordance with Drawing F10559699 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Environmental service conditions.

3.3.1 Extreme operating temperatures.- The mount shall meet the applicable requirements of 3.11 while exposed to, and thermally stabilized at, ambient temperatures of -40° and +150° Fahrenheit (F).

3.3.2 Standard ambient temperature.- Unless otherwise specified, the mount shall meet the following requirements while exposed to, and thermally stabilized at, a standard ambient temperature between +60° and +90°F.

3.3.3 Vibration.- The mount shall withstand a vibration of 30 ± 5 cycles per second (cps) frequency with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes ± 15 seconds. Subsequent to vibration the mount shall show no evidence of damage or physical failure, and it shall meet all requirements of this specification.

3.4 Orientation.- The mount shall be positioned as follows to meet the requirements of 3.5 to 3.10.3 inclusive. Where line of sight is specified, the requirement shall be met with the line of sight entering the telescope mounting sleeve from the end adjacent to the telescope elevation mechanism.

(a) The mounting dovetail surface in a vertical plane.

(b) The axis of the dovetail vertical.

(c) A line of sight provided which is parallel to the above mounting surface and which is perpendicular to the above axis.

(d) The elevation and cross level vial bubbles shall be centered within a half of one graduation space.

(e) The telescope elevation mechanism indices shall coincide within a half of the width of the graduation line.

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(f) The fixed coarse azimuth scale index indicating 3200 mils within a half of the width of the graduation line.

(g) The coarse elevation scale index indicating 800 mils within a half of the width of the graduation line.

(h) The indices for the graduated scales on the azimuth and elevation knobs indicating zero within 0.2 mil.

3.5 Excursion range.

3.5.1 Telescope elevation mechanism.- Rotation of the telescope elevation mechanism knob shall cause the line of sight to elevate and depress a minimum of 200 mils from the position specified in 3.4.

3.5.2 Elevation.- Rotation of the elevation knob shall cause the line of sight to elevate a minimum of 1000 mils and depress a minimum of 800 mils from the position specified in 3.4.

3.6 Plumb travel.

3.6.1 Telescope elevation mechanism.- Utilizing the telescope elevation mechanism knob, the line of sight shall track a plumb line within 0.5 mil when elevated and depressed through the range specified in 3.5.1 from the position specified in 3.4. The total lateral spread shall not exceed 0.5 mil.

3.6.2 Elevation.- Utilizing the elevation knob, the line of sight shall track a plumb line within 0.5 mil when elevated and depressed through the range specified in 3.5.2 from the position specified in 3.4. The total lateral spread shall not exceed 0.5 mil.

3.7 Lateral and vertical displacement.

3.7.1 Telescope elevation mechanism.- Lateral displacement of the line of sight shall not exceed 0.2 mil when the elevation motion is reversed by means of the knob.

3.7.2 Elevation.- Lateral displacement of the line of sight shall not exceed 0.2 mil when the elevation motion is reversed by means of the elevation knob.

3.7.3 Azimuth.- Vertical displacement of the line of sight shall not exceed 0.2 mil when the azimuth motion is reversed by means of the azimuth knob.

3.8 Horizontal travel.- The line of sight shall not vary in elevation in excess of 1.0 mil (total spread of 2 mils) when rotated 6400 mils in azimuth by means of the azimuth knob.

3.9 Angular accuracy.

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3.9.1 Open sight.- The line of sight of the open sight shall be parallel to the line of sight specified in 3.4 (c) within 10 mils.

3.9.2 Elevation.- The elevation scales shall indicate elevation angles within 0.7 mil, excluding backlash at each reading, when the line of sight is elevated and depressed through the range specified in 3.5.2. This requirement shall be met with the readings taken at 200 mil increments from the position specified in 3.4 (see 3.10.2).

3.9.3 Azimuth.- The fixed azimuth scales shall indicate azimuth deflection within 0.5 mil, excluding backlash at each reading, when the line of sight is rotated through 6400 mils in azimuth. This requirement shall be met with readings taken at 800 mil increments from the position specified in 3.4 (see 3.10.3).

3.10 Backlash.

3.10.1 Telescope elevation mechanism.- Backlash in the mechanism shall not exceed 0.5 mil.

3.10.2 Elevation.- When the amount of backlash at each elevation reading, taken in accordance with 3.9.2, is added to the elevation error at that setting, the sum shall not exceed 1.0 mil.

3.10.3 Azimuth.- When the amount of backlash at each azimuth reading, taken in accordance with 3.9.3, is added to the azimuth error at that setting, the sum shall not exceed 1.0 mil.

3.11 Operability.

3.11.1 Telescope clamping mechanism.- The telescope clamping mechanism shall be operable in the unlocked and locked positions when using a mandrel test piece having a diameter of .8427 - .0001 inches. These requirements shall be met at the standard ambient temperature of 3.3.2 and while the mount is subjected to the extreme operating temperatures of 3.3.1.

3.11.1.1 Unlocked position.- The mandrel test piece shall rotate freely and the clamping mechanism shall operate freely.

3.11.1.2 Locked position.- There shall be no movement of the mandrel test piece when a 40 inch pound torque is applied to the mandrel.

3.11.2 Mount locking latch.- The mount locking latch shall be operable at the standard ambient temperature of 3.3.2 and while subjected to the extreme operating temperatures of 3.3.1.

3.11.3 Locking knobs.- Subsequent to rotating the elevation and azimuth locking knobs to a locked position with a 10 pound-inch torque, there shall be no movement in their respective mechanisms when an 8 pound-inch torque is applied to the elevation and azimuth knobs. This requirement shall be met at the standard ambient temperature of 3.3.2 and while the mount is subjected to the extreme operating temperatures of 3.3.1.

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3.11.4 Azimuth slip scales.- The micrometer and coarse azimuth slip scales shall be operable at the standard ambient temperature of 3.3.2 and while subjected to the temperature requirements of 3.3.1.

3.11.5 Torque.- The running torque in inch ounces on all knobs shall be within the limits specified in Table I.

TABLE I

TORQUE VALUES (INCH OUNCES)

KNOB	+60° to 90°F (STANDARD TEMPERATURE) Min	Max	-40° and +150°F (EXTREME OPERATING TEMPERATURE) Min	Max
Telescope Elev. Mech.	10	20	10	40
Elevation	32	55	32	90
Azimuth	32	55	32	90

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three telescope mounts plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

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4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Table II herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE II

CLASSIFICATION CRITICAL: NONE MAJOR: AQL 1.0% Defective	REQUIREMENT	TEST PROCEDURE
101. Vibration	3.3.3	4.6.3, 4.6.3.1
102. Telescope elevation mechanism (excursion range)	3.5.1	4.6.4.1
103. Elevation (excursion range)	3.5.2	4.6.4.2
104. Telescope elevation mechanism (plumb travel)	3.6.1	4.6.4.3
105. Elevation (plumb travel)	3.6.2	4.6.4.4

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106.	Telescope elevation mechanism (displacement)	3.7.1	4.6.4.5
107.	Elevation (displacement)	3.7.2	4.6.4.6
108.	Azimuth (displacement)	3.7.3	4.6.4.8
109.	Horizontal travel	3.8	4.6.4.7
110.	Open sight (angular accuracy)	3.9.1	4.6.4.9
111.	Elevation (accuracy)	3.9.2	4.6.4.10
112.	Azimuth (angular accuracy)	3.9.3	4.6.4.11
113.	Telescope elevation mechanism (backlash)	3.10.1	4.6.4.12
114.	Elevation (backlash)	3.10.2	4.6.4.12
115.	Azimuth (backlash)	3.10.3	4.6.4.12
116.	Telescope clamping mechanism, unlocked position	3.11.1, 3.11.1.1	4.6.5, 4.6.5.1
117.	Telescope clamping mechanism, locked position	3.11.1, 3.11.1.2	4.6.5, 4.6.5.2
MINOR: 2.5% Defective			
201.	Mount locking latch (operability)	3.11.2	4.6.6
202.	Locking knobs (operability	3.11.3	4.6.7
203.	Azimuth slip scales (operability)	3.11.4	4.6.8
204.	Torque (operability)	3.11.5	4.6.9

NOTE: The tests in Table II shall be conducted at a standard ambient temperature of between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three mounts shall be selected at random as a special sampling from each 100 produced. The samples shall meet the requirements and tests in Table III.

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TABLE III

NO.	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
301.	Fabrication	3.1	Drawings-Visual
302.	General specification	3.2	MIL-F-13926-Visual

NOTE: The examinations in Table III shall be conducted at a standard ambient temperature of between 60°and 90°F.

4.4.2 Environmental tests.- Three telescope mounts shall be selected at random from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirement and test in Table IV and shall then meet the requirements for Characteristics 116, 117, 118, 201, 202, 203 and 204.

TABLE IV

NO.	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
303.	Low temperature operation (-40°F)	3.3.1, 3.11	4.6.1
304.	High temperature operation (+150°F)	3.3.1, 3.11	4.6.1

4.4.3 Failure of sample.- Should any item of a special sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- When the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance of

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which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.-

4.5.2.3.1 Orientation.- Orientation in conformance with the requirements of 3.4 is accomplished by using the elevation and cross leveling testing fixture, Drawing D6523553. The fixture gage assembly, shown on Drawing D6510966, shall be assembled to the cross leveling fixture. An auxiliary telescope shall be placed on the V's, and the cross leveling fixture shall be leveled with a 10-second level for both directions in the horizontal plane. The auxiliary telescope shall be sighted on a wall target or in a target collimator by moving the fixture gage assembly. Place the perpendicularity gage, shown on Drawing D7646979, in the dovetail slot of the fixture gage assembly. Place the auxiliary telescope in the "V" slot and adjust for zero line of sight in the vertical position.

4.5.2.3.2 Wall target.- A calibrated wall target capable of measuring and providing the angular mechanical accuracies of the mount shall be located approximately fifty feet from the elevation and cross leveling fixture.

(a) The target shall contain a "0" horizontal line of sight, with the elevation and depression images positioned in the same vertical plane with the "0" horizontal line of sight. The angular position of the lines of sight of the elevation and depression target images shall be established in mils at elevation and depression settings of +200, +1000, -200 and -800.

(b) The wall target shall contain a calibrated vertical image line (plumb line) for measuring travel accuracy of the elevation test fixture prior to performing accuracy tests of the telescope mount. The wall target shall also contain calibrated displacement marks with respect to the vertical image line (plumb line) for measuring plumb travel. Tolerance marks shall be incorporated on both sides of this vertical plumb line to conform with the tolerances specified in 3.6.1 and 3.6.2.

(c) The perpendicularity gage shall be positioned in the slot of the elevation fixture that is perpendicular to its axis. A collimating telescope of approximately six-power with a suitable cross line reticle shall be utilized. The reticle shall be centered with respect to the outside diameter within 10 seconds. Provisions shall be made to permit plumbing the reticle within 10 seconds and for maintaining this accuracy with respect to the perpendicularity gage and with the target. Positioning the collimating telescope in the Vee slot of the perpendicularity gage will establish a line of sight through the telescope mounting sleeve.

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4.6 Test methods and procedures.

4.6.1 Extreme operating temperatures.- The mount shall be exposed to the temperature conditions specified in 3.3.1, utilizing testing equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. The mounts shall be placed in the cold chamber and the temperature of the chamber reduced gradually -40°F. The mounts shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, and while at -40°F the mounts shall be subjected to the tests in 4.6.5 to 4.6.9 inclusive to determine compliance with 3.3.1. Upon completion of this test the mount temperature shall be gradually raised to room ambient temperature. The mount shall then be placed into an oven chamber and the temperature shall be raised gradually to +150°F. The mounts shall remain at this temperature for a period of 5 hours and at the conclusion of this 5 hour period, and while at +150°F, the mounts shall again be subjected to the tests in 4.6.5 to 4.6.9 inclusive to determine compliance with 3.3.1. Upon completion of this test the temperature of the test chamber shall be gradually reduced to room ambient temperature (60° to 90°F) at which time the mounts shall be removed from the test chamber. Caution shall be exercised during this test to avoid subjecting the mounts to thermal shock.

NOTE: The rate of temperature change in the test chamber shall not exceed 20°F per hour throughout the temperature cycling test of 4.6.1.

4.6.2 Standard ambient temperature.- The mount shall pass the following tests while thermally stabilized at an ambient temperature between +60° and +90°F as specified in 3.3.2 and the "Test Facilities" of MIL-F-13926.

4.6.3 Vibration.- Standard vibration equipment capable of providing the amplitude and frequency as specified in 3.3.3 shall be utilized for this test. The mount shall be vibrated in a vertical plane. Unless otherwise specified, the test equipment shall be in conformance with the "Test Facilities" of MIL-F-13926.

4.6.3.1 Post vibration.- Subsequent to vibration the mount shall be throughly examined for physical defects resulting from poor workmanship and material to determine compliance with 3.3.3 prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts prior to being subjected to the remaining performance tests. Minimum visual examination shall include the level vials (also use smell tests to detect a leaking vial resulting from a broken seal or glass), screws, nuts, etc. A tactile examination shall also be conducted to determine looseness or improper functioning of the elevation, azimuth, and angle of site knobs, the clamping mechanism, mounting latch, locking knobs,and azimuth slip scales. Any one of the possible defects defined above shall be cause to consider a mount defective. Mounts which have passed the above examinations shall be tested for all other requirements of this specification.

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4.6.4 Orientation.- Perform the following tests with the telescope mount positioned on special testing equipment conforming to 4.5.2.3.1.

4.6.4.1 Telescope elevation mechanism, excursion.- Use the telescope elevation mechanism knob to elevate and depress a minimum of 200 mils from the zero position established in 4.5.2.3.1. While observing through the collimating telescope, the line of sight of the telescope shall meet the requirements of 3.5.1 as measured on the target image.

4.6.4.2 Elevation, excursion.- Observe through the collimating telescope, wherein the center point of the telescope reticle shall be in coincidence with respect to the intersecting point of the "0" (horizontal line of sight) and plumb line of the target image. Utilize the elevation knob to elevate the telescope a minimum of 1000 mils. In cases where the plumb line is not long enough for full excursion, the elevation fixture can compensate for full travel for the test of 3.5.2 by rotating the elevation fixture. The horizontal line of the telescope reticle shall be in coincidence with respect to the 1000 mil horizontal target image line. From zero position, rotate the elevation knob to depress the line of sight of the telescope a minimum of 800 mils. The horizontal line of the telescope reticle shall be in coincidence with respect to the 800 mil horizontal target image line to ascertain conformance with the requirements of 3.5.2.

4.6.4.3 Telescope elevation mechanism, plumb travel.- Sight through the collimating telescope and observe that the plumb line is in coincidence with the center point (horizontal and vertical lines) of the telescope reticle (see 4.5.2.3.1). Rotate the telescope elevation mechanism knob through the excursion specified in 3.5.1. During this rotation the vertical line of the telescope reticle shall be in coincidence with respect to the vertical target (plumb line) image line within the tolerance specified in 3.6.1. The total deviation between any two error readings shall not exceed the tolerance specified in 3.6.1.

4.6.4.4 Elevation, plumb travel.- The test equipment defined in 4.5.2.3.1 and the test methods as depicted in the above paragraphs shall apply, except that the elevation knob shall be utilized for this test. The line of sight shall track the plumb line in accordance with 3.6.2. The total deviation between any two error readings shall not exceed the tolerance specified in 3.6.2.

4.6.4.5 Telescope elevation mechanism displacement.- This test may be performed in conjunction with 4.6.4.4, using the same test equipment and set-up. Actuate the telescope elevation mechanism knob and note the position of the line of sight to the mil block chart, then reverse the direction. Any lateral displacement of the line of sight shall not exceed the tolerance specified in 3.7.1.

4.6.4.6 Elevation displacement.- Use the same test equipment and test methods as in 4.6.4.5, except that the elevation knob shall be used instead of the telescope elevation mechanism knob for this test. Any lateral displacement of the line of sight shall not exceed the tolerance of 3.7.2.

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4.6.4.7 Horizontal travel.- The horizontal travel test is performed with the aid of the azimuth test fixture shown on Drawing F7691596, the collimating telescope, and adapter assembly. The adapter assembly is attached to the mounting plate of the azimuth test fixture and provides a slot for positioning the mount. With the azimuth fixture crossleveled, the slot is positioned in a vertical plane with the perpendicularity gage. The test consists of checking the line of sight of the telescope mounting sleeve for deflection at 0,1600, 3200, and 4800 mil settings of the azimuth test fixture during a 6400 mil excursion of the mount in a horizontal plane. The error shall not exceed the requirements of 3.8.

4.6.4.8 Azimuth displacement.- This test may be combined with 4.6.4.7 above. A check for any vertical displacement shall be made at the four positions of the azimuth test fixture as described in 4.6.4.7. Vertical displacement of the line of sight shall not exceed the tolerance as specified in 3.7.3.

4.6.4.9 Open sight.- The open sight test is made by means of a wall chart located approximately fifty feet from the telescope mount. The chart shall have a mark for the line of sight through the telescope mounting sleeve, and a ten-mil radius circle the center of which will correspond to the center of the open sight. With the aid of the collimator focused on the line of sight mark, the open sight shall read within the limits as specified in 3.9.1.

4.6.4.10 Elevation accuracy.- Elevation accuracy is checked using the same set-up as used for the plumb travel test of 4.6.4.4. In addition, a horizontal zero mark shall be located on a wall chart in back of the plumb line and in a horizontal plane with the line of sight through the telescope mounting sleeve. A mil block located above the line and one below will indicate the error for this test. With the mount located in the dovetail of the fixture, the cross lines of the collimator shall coincide with the zero mark of the wall chart. Elevate the fixture to 11 degrees 15 minutes (200 mils). Then rotate the elevation knob of the mount 200 mils. This should coincide the cross lines of the collimator with the horizontal mark of the wall target. Deviation can be read on the mil blocks and shall be within the tolerance of 3.9.2. This operation shall be repeated for the entire scale reading at 200 mil increments.

4.6.4.11 Azimuth accuracy.- The azimuth accuracy test is made with the same set-up employed in the horizontal travel test of 4.6.4.7. With the mount and azimuth fixture set at zero and in alignment with the target collimator of the fixture, readings are taken in 800 mil increments through 6400 mils of travel and the deviation shall not exceed the requirements of 3.9.3.

4.6.4.12 Backlash.- Backlash shall be checked at positions where accuracy readings are taken on the targets described for these tests. The backlash test consists of sighting on the target through a collimating telescope installed on the mount, clockwise movement of each micrometer knob without over-travel, and then a counterclockwise movement without over-travel. The difference in readings of each movement shall be considered backlash. The tolerance of these tests shall not exceed the requirements of 3.10.1, 3.10.2, and 3.10.3.

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4.6.5 Telescope clamping mechanism, operability.- The telescope clamping mechanism is tested manually for operability at temperatures specified in 3.3.1 and 3.3.2. When clamped, the test piece as specified in 3.11.1 shall be firmly held in all positions when applying the specified torque, using a commercial torque wrench. Upon releasing the clamping mechanism, there shall be free movement in all directions.

4.6.5.1 Unlocked position.- When the specified test piece is positioned in the clamping mechanism, it shall rotate freely without drag or interference as specified in 3.11.1.1 It shall meet this requirement when subject to the temperatures specified in 3.3.1 and 3.3.2".

4.6.5.2 Locked position.- With the specified test piece positioned in the clamping mechanism as in 4.6.5.1, the clamping mechanism shall be actuated to the locked position. The test piece shall have no movement when the specified torque (3.11.1.2) is applied. The clamping mechanism shall operate freely when subjected to temperatures specified in 3.3.1 and 3.3.2".

4.6.6 Mount locking latch, operability.- The locking latch shall be manually operable while subjected to the temperatures specified in 3.3.1 and 3.3.2 to determine compliance with 3.11.2.

4.6.7 Locking knobs, operability.- Locking action of the elevation and azimuth locking knobs shall be checked at the temperatures specified in 3.3.1 and 3.3.2 for compliance with the requirements of 3.11.3, utilizing the commercial torque wrench with an adapter.

4.6.8 Azimuth slip scales, operability.- The micrometer and coarse azimuth slip scales shall be examined by manually slipping the scales while subjected to the temperatures specified in 3.3.1 and 3.3.2 to determine compliance with 3.11.4.

4.6.9 Torque.- This test shall be performed using a commercial type inch-ounce torque wrench with adapters to accommodate the knobs. Readings taken with the torque wrench shall be taken in both directions and while in motion. The torque required to rotate the knobs as specified in Table I at standard ambient temperature shall not be exceeded by more than the requirement of 3.11.5 when tested at the extreme operating temperatures of 3.3.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10559699. The level of protection shall be as specified in the procurement document. (see 6.2)

6. NOTES

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6.1 Intended use.- Telescope Mount: M128A1 is a component of Sightunit: M53A1. The mount introduces measured azimuth and elevation readings to Elbow Telescope: M109. The sightunit is primarily used to lay mortars and rocket launchers for indirect fire.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selction of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).
(e) Provisions for First Article testing.

Custodian:

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Preparing activity:

Army - MU

Project No. 1240-A209

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 65 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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Department of the Army

Frankford Arsenal
Philadelphia, Pa. 19137

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Department of the Army

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-N1100
Philadelphia, Pa. 19137

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